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Dendritic growth and synaptic organization from activity-independent cues and local activity-dependent plasticity

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Abstract

Dendritic branching and synaptic organization shape single neuron and network computations. How they emerge simultaneously during brain development as neurons become integrated into functional networks is still not mechanistically understood. Here, we propose a mechanistic model in which dendrite growth and the organization of synapses arise from the interaction of activity-independent cues from potential synaptic partners and local activity-dependent synaptic plasticity. Consistent with experiments, three phases of dendritic growth – overshoot, pruning, and stabilization – emerge naturally in the model. The model generates stellate-like dendritic morphologies capturing several morphological features of biological neurons under normal and perturbed learning rules, reflecting biological variability. Model-generated dendrites have approximately optimal wiring length consistent with experimental measurements. Besides setting up dendritic morphologies, activity-dependent plasticity rules organize synapses into spatial clusters according to the correlated activity they experience. We demonstrate that a trade-off between activity-dependent and -independent factors influences dendritic growth and synaptic location throughout development, suggesting that early developmental variability can affect mature morphology and synaptic function. Therefore, a single mechanistic model can capture dendritic growth and account for the synaptic organization of correlated inputs during development. Our work suggests concrete mechanistic components underlying the emergence of dendritic morphologies and synaptic formation and removal in function and dysfunction, and provides experimentally testable predictions for the role of individual components.

eLife assessment

This **important** work investigates how two distinct processes, morphological changes and synaptic plasticity, contribute to the final shape of neuronal dendrites and the spatial structure of their synaptic inputs. The modelling is **solid** and could be broadly applied to other similar questions. Some of the findings are a direct result of how the model is constructed, which limits the impact. The work will be of interest to neuroscientists studying dendritic development and connectivity at a single-cell level.

Introduction

The dendrites of a neuron are intricately branched structures that receive electrochemical stimulation from other neurons. The morphology of dendrites determines the location of synaptic contacts with other neurons and thereby constrains single-neuron computations. During development, the dendrites of many neurons grow simultaneously and become integrated into neural circuits. Dendrite development is highly dynamic; iterated addition and retraction of branches allow these dendrites to probe various potential synaptic partners before stabilizing ((Cline, 2016); (Richards et al., 2020)). Many intrinsic and extrinsic factors underlie the dynamics of dendritic development. In any given neuron, intrinsic expression of specific genes controls many morphological aspects, including the orientation of the dendrite in the cortex, the general abundance of dendritic branching, and the timing of growth onset ((Puram and Bonni, 2013)). Extrinsic signaling, in contrast, exerts precise control over the detailed dynamics of dendrite development via various mechanisms, including activity-dependent cell-to-cell interactions and molecular signaling ((Polleux et al., 2016)).

While many signaling molecules affect dendrite development, the brain-derived neurotrophic factor (BDNF) and its immature predecessor proBDNF are particularly crucial in the central nervous system ((Lu et al., 2005)). While exposure to BDNF leads to larger dendrites with a higher density of synapses ((McAllister et al., 1995); (Tyler and Pozzo-Miller, 2001)), exposure to proBDNF leads to smaller dendrites with fewer synapses ((Koshimizu et al., 2009); (Yang et al., 2014)). Furthermore, the precise balance of BDNF and proBDNF is essential for the organization of synapses into clusters during development ((Kirchner and Gjorgjieva, 2021); (Winnubst et al., 2015); (Kleindienst et al., 2011); (Niculescu et al., 2018)). Interestingly, synaptic activity triggers the cleaving of proBDNF into BDNF ((Je et al., 2012)), providing a mechanistic link between the molecular factors driving dendrite maturation and neural activity.

Activity-dependent factors are equally important in driving dendritic growth. As the sensory periphery is immature during early postnatal development, when many dendrites grow ((Leighton and Lohmann, 2016)), many developing circuits generate their own spontaneous activity. The rich spatiotemporal structure of spontaneous activity instructs the formation, removal, and change in the strength of synaptic inputs ((Sretavan et al., 1988); (Sakai, 2020)) and triggers the stabilization or retraction of entire dendritic branches ((Riccomagno and Kolodkin, 2015); (Lohmann et al., 2002)). While blocking spontaneous activity does not result in grossly different dendrite morphology, the density and specificity of synaptic connections are strongly perturbed ((Campbell et al., 1997); (Ultanir et al., 2007)), highlighting the instructive effect of spontaneous activity on dendritic development ((Crair, 1999)).

One influential hypothesis tying together the extrinsic signaling factors underlying dendritic development is the synaptotrophic hypothesis ([\(Vaughn, 1989\)](#)). According to this hypothesis, a growing dendrite preferentially extends into regions where it is likely to find synaptic partners. Once a dendrite finds such a partner, a synaptic contact forms, anchors the developing dendrite, and serves as an outpost for further dendritic growth. Conversely, loss of synaptic input to the dendrite can lead to retraction unless the remaining synapses stabilize the branch ([\(Lohmann et al., 2002\)](#); [\(Niell et al., 2004\)](#); [\(Haas et al., 2006\)](#); [\(Cline and Haas, 2008\)](#); [\(Riccomagno and Kolodkin, 2015\)](#); [\(Cline, 2016\)](#)). However, elaborate dendrites with morphologically defined synapses can also emerge without any synaptic transmission ([\(Verhage et al., 2000\)](#); [\(Cijssouw et al., 2014\)](#)), suggesting that synaptic activity influences dendritic growth but is not the only driving force. Despite the significant interest in the synaptotrophic hypothesis, we still lack a mechanistic understanding of how activity-dependent and -independent factors combine to shape development.

To investigate interactions between known signaling factors and to synthesize information from different experimental results, computational models of dendrite development provide a fruitful direction to explore how different mechanisms can generate realistic dendritic morphologies ([\(Cuntz, 2016\)](#)). Previous approaches include modeling dendritic development with random branching ([\(Kliemann, 1987\)](#)) or as a reaction-diffusion system ([\(Luczak, 2006\)](#)), implementing activity-independent growth cones that sense molecular gradients ([\(van Veen and van Pelt, 1992\)](#); [\(Torben-Nielsen and De Schutter, 2014\)](#)), or constructing dendrites as the solution to an optimal wiring problem ([\(Cuntz et al., 2010\)](#)). While these approaches can generate dendrites that accurately match the statistics of developing and mature biological dendrites ([\(Koene et al., 2009\)](#); [\(Cuntz, 2016\)](#)), they provide limited insight into how dendritic growth interacts with synapse formation and local activity-dependent organization of synaptic inputs, hence obscuring the link between morphological variability and electrophysiological ([\(Gouwens et al., 2020\)](#); [\(Scala et al., 2021\)](#)) or functional ([\(Poirazi and Mel, 2001\)](#); [\(Poirazi et al., 2003\)](#); [\(Park et al., 2019\)](#); [\(Poirazi and Papoutsis, 2020\)](#)) synaptic and dendritic properties.

Here, we propose a mechanistic computational model for cortical dendritic development for dendrite growth and synapse formation, stabilization and elimination based on reciprocal interactions between activity-independent growth signals and spontaneous activity. Starting from neuronal somata distributed in a flat sheet of cortex, spatially distributed potential synapses drive the growth of stellate-like dendrites through elongation and branching by activity-independent cues. Upon contact, synaptic connections form and stabilize or disappear according to a local activity-dependent learning rule inspired by neurotrophin interactions based on correlated patterns of spontaneous activity ([\(Kirchner and Gjorgjieva, 2021\)](#)). Consistent with the synaptotrophic hypothesis, the stability of a dendritic branch depends on the stability of its synaptic contacts, with the branch likely retracting after substantial synaptic pruning. The resulting dynamic system naturally leads to the emergence of three distinct phases of dendrite development: 1) an initial overshoot phase characterized by dendrite growth and synapse formation, 2) a pruning phase during which the learning rule prunes poorly synchronized synapses, and 3) a stabilization phase during which morphologically stable dendrites emerge from the balancing of growth and retraction. Varying model parameters in biologically realistic ranges produces dendrite length and synapse density changes consistent with experiments. Our mechanistic model generates dendrites with approximately optimal wiring length, which is a widely used criterion for evaluating dendritic morphology ([\(Cuntz et al., 2010\)](#), [\(2012\)](#); [\(Chklovskii et al., 2002\)](#)). At the same time, the model leads to the activity-dependent emergence of functional synaptic organization and input selectivity. Therefore, our mechanistic modeling framework for the growth and stabilization of dendritic morphologies and the simultaneous synaptic organization according is ideally suited for making experimental predictions about the

effect of perturbing specific model components on the resulting dendritic morphologies and synaptic placement.

Results

We built a computational model of activity-dependent dendrite growth during development based on synapse formation, stabilization, and elimination. We focused on basal stellate-like dendrites of cortical pyramidal neurons, which primarily extend laterally within a layer of the cortex ((*Larkman and Mason, 1990*)) and receive numerous feedforward and recurrent inputs ((*Rossi et al., 2019*); (*Iacaruso et al., 2017*)). Stellate morphologies are found in many types of neurons, especially in the somatosensory cortex, including interneurons and layer 4 spiny stellate cells, which are the main recipients of thalamic inputs and play a key role in sensory processing ((*Schubert et al., 2003*); (*Marques-Smith et al., 2016*); (*Scala et al., 2019*)). To investigate the impact of synapse formation on dendrite development, we modeled several neuronal somata and potential synapses in a flat sheet of cortex (*Figure 1a*). Potential synapses represent locations in the cortex where an axon can form a synapse with a nearby dendrite ((*Stepanyants and Chklovskii, 2005*)). The model consists of two components: An activity-independent component that directly governs branch growth and retraction; and an activity-dependent component that governs synaptic organization and thus indirectly affects branch stability. Inspired by the synaptotrophic hypothesis ((*Vaughn, 1989*)), we mimicked the effect of activity-independent molecular signaling by letting each potential synapse release diffusive signaling molecules that attract the growing dendrite (*Figure 1b*, *Figure 1–Figure Supplement 1*, *Figure 1–Figure Supplement 2*). In addition, during development and before the onset of sensory experience, neural circuits generate patterned spontaneous activity ((*Blankenship and Feller, 2010*); (*Ackman and Crair, 2014*)). Therefore, to model the structured spontaneous activity exhibited by different axons ((*Scholl et al., 2017*); (*Iacaruso et al., 2017*)), we divided potential synapses randomly into different activity groups that receive inputs correlated within a group but uncorrelated between groups (see Methods). Each group represents either synapses from the same presynaptic neuron or from neurons that experience correlated presynaptic activity.

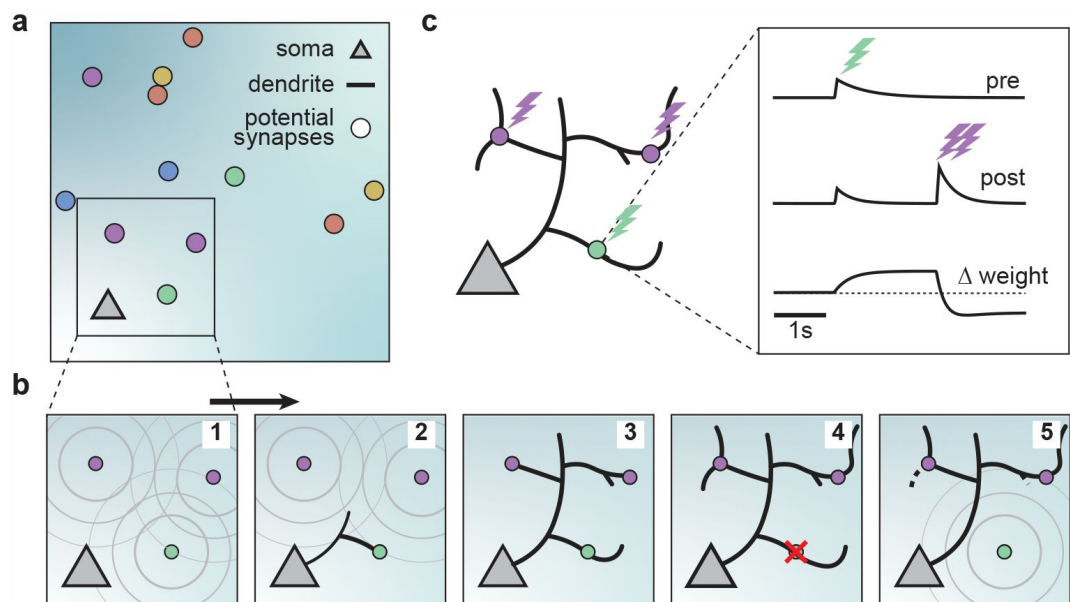


Figure 1.

A model of dendritic growth for a cortical pyramidal neuron driven by activity-independent and -dependent mechanisms.

(a) Schematic of the soma of a pyramidal neuron (orange triangle) with 12 randomly distributed potential synapses from presynaptic axons (circles) with correlated activity patterns indicated by color. (b) Schematic of activity-independent and -dependent mechanisms. Soma and synapses correspond to box in a. Signaling molecules diffusing from potential synapses (1) attract dendrite growth and promote synapse formation (2) independent of firing pattern (3). Over time, poorly synchronized synapses depress and are pruned from the dendrite (4), while well-synchronized synapses remain stable (5). After a branch retracts, the dendrite is less sensitive to the growth field at that location (5). (c) Change in weight of one synapse (green) following the stimulation of itself (green bolt) and of two nearby synapses (purple bolts). Left: Schematic of the developing dendrite from b with bolts indicating synaptic activation. Right: Presynaptic accumulator (top), postsynaptic accumulator (middle), and change in synaptic weight (bottom) as a function of time (see Methods [Kirchner and Gjorgjieva \(2021\)](#) for details of the plasticity rule). Dashed line (bottom) indicates zero change.

Figure 1–Figure supplement 1. The growth field is similar to two-dimensional heat diffusion.

Figure 1–Figure supplement 2. Detailed illustration of the dendritic growth mechanism.

Because of their attraction to growth-factor releasing synapses and independent of neural activity, dendrites in our model grow outward from the soma towards the nearest potential synapse, where they form a synapse and temporarily stabilize ([Figure 1b](#), [Figure 1–Figure Supplement 2](#)). We assumed that dendrites could not overlap based on experimental data ([Grueber and Sagasti, 2010](#)); therefore, dendrites in the model retract, for instance, when further growth would require self-overlap. Once a synapse is formed, we modeled that its strength changes according to a local, activity-dependent plasticity rule ([Kirchner and Gjorgjieva, 2021](#)) ([Figure 1c](#)). The learning rule induces synaptic potentiation whenever presynaptic and local postsynaptic activity co-occur, and synaptic depression whenever local

postsynaptic activity occurs in the dendrite independent of presynaptic stimulation, usually due to the activation of a neighboring synapse (see Methods and the ‘offset’ constant below),

$$\Delta \text{weight} = \text{post} \times (\text{pre} - \text{offset}). \quad (1)$$

As shown previously, this rule generates synaptic distance-dependent competition, where nearby synapses affect each other more than distant synapses, and correlation-dependent cooperation, where neighboring synchronized synapses stabilize. In contrast, neighboring desynchronized synapses depress (*Kirchner and Gjorgjieva, 2021*). In our model, we assumed that when a synapse depresses below a threshold, it decouples from the dendrite, and the corresponding branch retracts successively to either the nearest stable synapse, branch point, or the soma (*Figure 1b, Figure 1–Figure Supplement 2*). After removal, the vacated synapse turns into a potential synapse again, attracting other growing branches. Thus, a developing dendrite in our model acquires its arborization through the attraction to signaling molecules released by potential synapses and the repeated activity-dependent formation, stabilization and removal of synapses.

Dendrite development through balancing growth and retraction

After specifying the rules governing the growth of individual dendritic branches, we investigated dendritic development on long timescales. When growing dendrites according to our proposed growth rule based on signaling molecules attraction and spontaneous activity-dependent synaptic refinements (*Figure 1*), we found that dendrites form several stems, i.e. branches which start directly at the soma, and rapidly expand outwards (*Figure 2a*). After an initial phase of rapid expansion, we observed that growth rapidly attenuates, and the dendritic length stabilizes (*Figure 2b*). This stability is achieved when the dendrite’s expansion and retraction are balanced (*Figure 2c*). To investigate whether the stability in total length also corresponds to stability in dendritic morphology, we quantified morphological stability as the pixel-wise correlation of a dendrite with itself 4.5 hours earlier, which is several orders of magnitude larger than the speed at which dendrites grow and retract in our model (see *Table 1*). Despite the residual amount of expansion and retraction, we showed that morphological stability increases rapidly, and the dendritic morphology is already stable after the initial expansion phase (*Figure 2d*). Interestingly, such rapid stabilization of morphology has also been observed in the mouse visual cortex (*(Richards et al., 2020)*) and the *Drosophila* larvae (*(Castro et al., 2020)*). We next quantified the Sholl diagram, the number of dendritic branches at a given distance from the soma, commonly used as a measure of dendritic complexity (*(Sholl, 1953)*; *(Binley et al., 2014)*; *(Bird and Cuntz, 2019)*). The Sholl diagram of the stabilized dendrites generated by our model is unimodal and qualitatively matches the Sholl diagram of developing basal dendrites from the mouse medial prefrontal cortex (*Figure 2e*; data extracted from ref. *(Kroon et al., 2019)*, postnatal days 6–8), as well as the hippocampus (*(Kleindienst et al., 2011)*). In summary, by combining activity-independent and -dependent dendritic growth mechanisms, our model produces dendrites that rapidly expand and stabilize by balancing growth and retraction.

Table 1.

Parameters of the minimal plasticity model (*(Kirchner and Gjorgjieva, 2021)*) and the synap-totrophic growth model.

Parameter	Variable	Value
Synaptic efficacy time constant	τ_W	6000 time steps
Postsynaptic accumulator time constant	τ_u	300 time steps
Presynaptic accumulator time constant	τ_v	600 time steps
Constitutive percent of BDNF of total neurotrophins	η	45%
MMP9 efficiency constant	ϕ	$\frac{3}{50}$ per time step
Heterosynaptic offset	ρ	$\rho = \frac{2\eta-1}{2(1-\eta)}$
Minimal model synaptic efficacy time constant	τ_w	$\tau_w = \tau_W \frac{1}{2(1-\eta)}$
Standard deviation of calcium spread	σ_c	200 μm
Turnover threshold below which a synapse is replaced	W_{thr}	0.02
Firing rate of synapses	r_{in}	0.116 min^{-1}
Scout intervals and speed	$t_{\text{scout}}, v_{\text{scout}}$	10 min, 0.18 $\mu\text{m min}^{-1}$

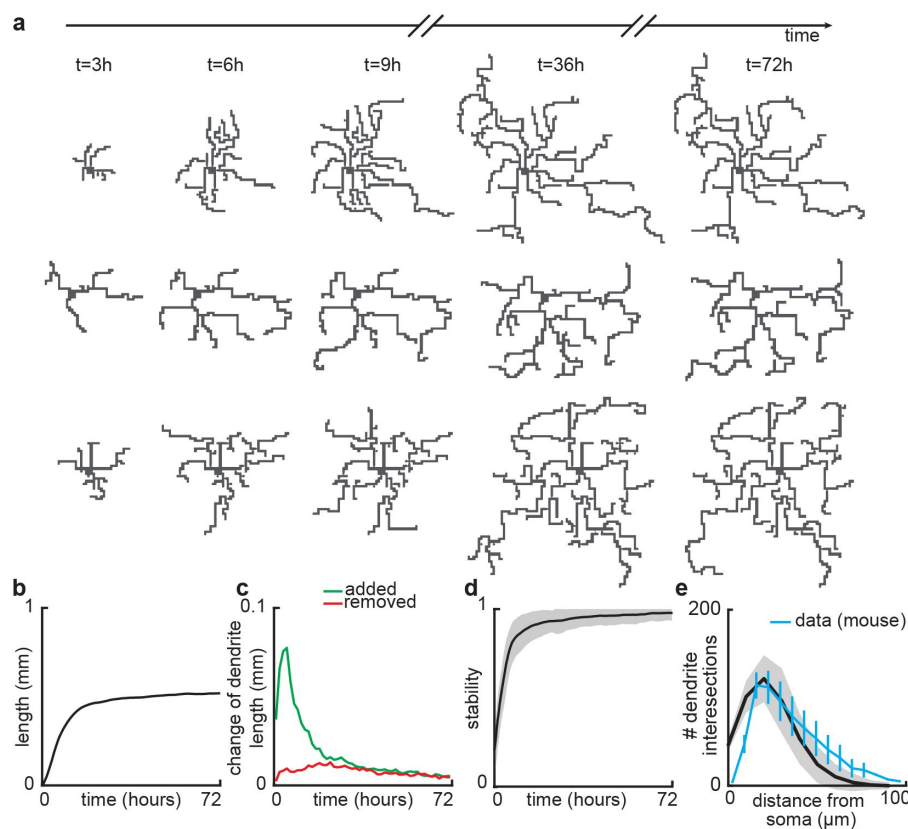


Figure 2.

Balanced growth and retraction generate morphologically stable dendrites.

(a) Three example dendrites at five time points from our simulations. For clarity of presentation, connected synapses are not displayed. (b) Total length of dendritic tree as a function of time. (c) Length of dendrite added (green) and removed (red) as a function of time. (d) Morphological stability (correlation between the dendrite shape at time t and $t - 4.5$ hours) as a function of time. (e) Average number of dendrite intersections as a function of distance from the soma (the Sholl diagram). Data from basal dendrites in the developing mouse medial prefrontal cortex superimposed, normalized to the maximum (blue; ref. ((Kroon *et al.*, 2019))). All lines

represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations.

Figure 2-video 1. Example of a simulation in which several dendrites develop in parallel.

Delayed activity-dependent plasticity produces a rapid increase of synapse density followed by pruning

Since our model couples dendritic growth to the formation and removal of synapses (Figure 3a), we next investigated how the number of connected synapses, which are necessary for the dendrite's stabilization, changes over time. As a result of the dendrite's rapid growth, we observed a rapid increase in the number of connected synapses (Figure 3b,c). In contrast to the dendrite's length, we found that the initial rapid increase in connected synapses is followed by a brief period of an over-all reduction of the number of synapses before additions and removals are balanced (Figure 3c). This removal of established synapses resembles the postnatal removal of synapses observed in the mouse neocortex ((Holtmaat *et al.*, 2005)). To understand how the initial overshoot and subsequent removal of synapses emerge in our model, we computed the average synaptic weight of all synapses that eventually stabilize or are pruned (Figure 3d). We found that the delayed onset of synapse removal (Figure 3c) is due to the slow time scale of the synaptic weight compared to the faster time scale of dendrite growth. Thus, the initial overshoot and subsequent removal of synapses observed in our model (Figure 3b) is due to the rapid formation relative to the delayed activity-dependent elimination of synapses.

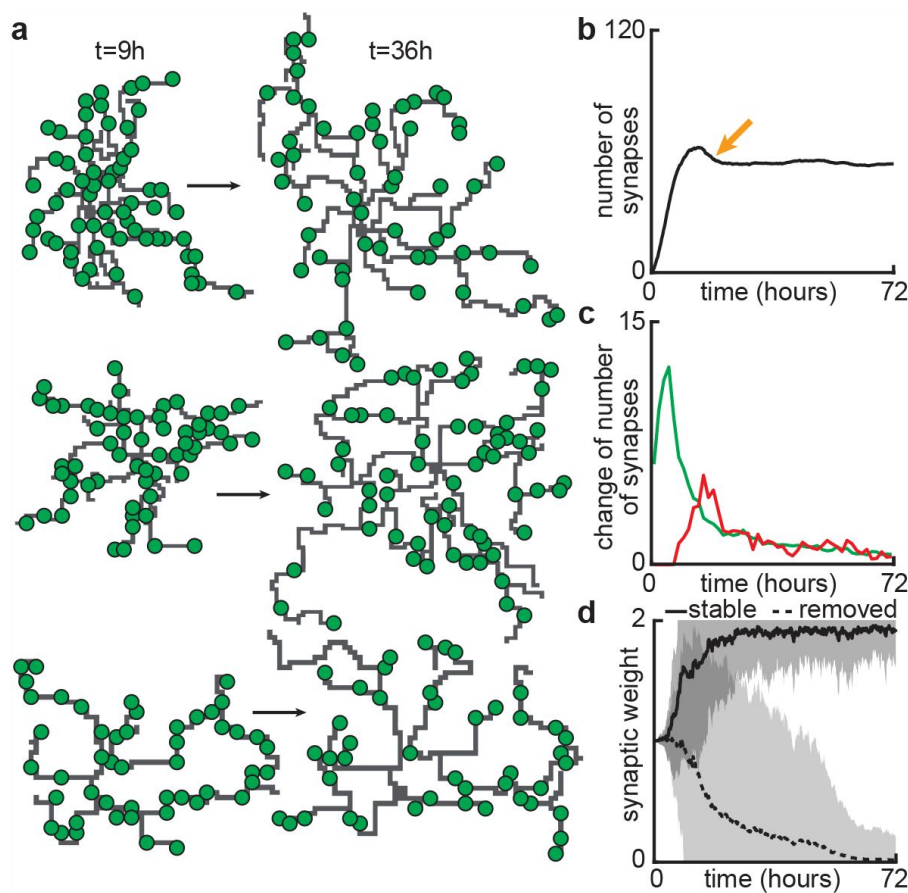


Figure 3.

Synapse formation and removal predominate in distinct phases of dendrite development.

(a) Three examples of dendrites at the beginning ($t = 9$ hours) and end ($t = 72$ hours) of the simulation. Green circles indicate formed synapses. (b) Total number of connected synapses as a function of time. Orange arrow highlights overshoot and subsequent pruning. (c) Added (green) and pruned synapses (red) as a function of time. (d) Average synaptic weights of synapses that ultimately stabilize (solid black; final weight more than 0.5) or are removed (dashed black; final weight less than 0.5) as a function of time. All lines represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations.

Activity-dependent competition between synapses produces input selectivity and synaptic organization

Next, we asked if the stabilization of dendrites might be supported by the emergence of organization of connected synapses. First, we compared the synapses connected to dendrites at the apex of the overshoot phase (peak in [Figure 3b](#)) with those in the stabilization phase ([Figure 4a,b](#)). While dendrites at the apex do not prefer synapses from any particular input group, in the stabilization phase, they acquire a preference for synapses from only two or three of the activity groups ([Figure 1b](#)). These dynamics resemble the activity-dependent synaptic competition in the developing visual cortex, where asynchronously activated synapses tend to depress ([Winnubst et al., 2015](#)). Notably, the remaining synchronized synapses in our model experience correlation-dependent cooperation ([Kirchner and Gjorgjieva, 2021](#)), and are therefore able to stabilize the dendrite and prevent the total retraction of all branches.

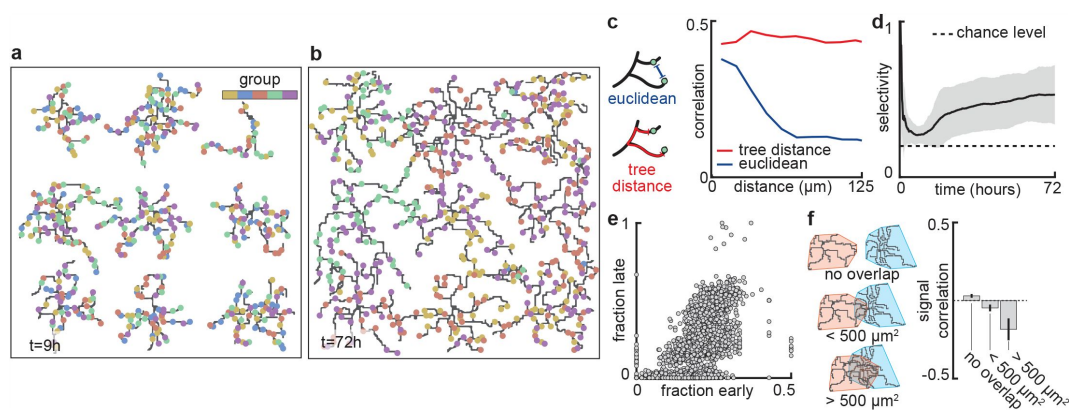


Figure 4.

Stable morphology is obtained through selective removal of synapses and dendritic input selectivity.

(a,b) Dendritic trees before (a, 9 hours) and after (b, 72 hours) removal of synapses (Figure 3). Connected synapses colored corresponding to activity group, which represents activity correlations (Figure 1b). (c) Left: Schematic illustrating the difference between Euclidean and tree distance. Note that we compute the Euclidean distance between synapses from different trees. Right: Correlation between pairs of synapses as a function of the Euclidean distance (blue) and tree distance (red). (d) Input selectivity of dendrites (defined as the fraction of the activity group with the highest representation) as a function of time. Dashed line indicates chance level. All lines represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations. (e) Fraction of connected synapses per activity group early ($t = 9$ hours) and late ($t = 72$ hours) in the simulation. Each dot represents one of the five activity groups on one of the nine dendrites from the 32 simulations, resulting in $5 \times 9 \times 32 = 1440$ data points. (f) Left: Schematic of different levels of overlap (rows) between the convex hulls of two dendrites, referring to the smallest convex sets that contain the dendrite. Right: Signal correlation (correlation between fractions of synapses from the same activity groups) for different levels of dendritic overlap. Error bars indicate the standard error of the mean, computed from 1152 pairs of dendrites from 32 simulations.

This selective potentiation of synapses according to input correlation also leads to dendritic selectivity for inputs. In particular, synapses on the same dendrite are likely to come from the same activity group (Figure 4c). This selectivity is acquired throughout the simulation, where selectivity starts high (a nascent dendrite is trivially selective to its first synapse; $t = 0-1$ hours), drops almost to chance level (indiscriminate addition of synapses; $t = 9$ hours), and finally approaches a value of $\frac{1}{2}$ (two activity groups per dendrite remain after the pruning phase; $t = 72$ hours) (Figure 4d). To determine which activity group stabilizes eventually, we computed selectivity for each group early ($t = 9$ hours) and late ($t = 72$ hours). We found that early high (low) selectivity for an activity group translates into even higher (lower) selectivity in the stable state (Figure 4e), predicting an outsized impact of early synaptic contacts on continued dendritic function. Finally, we observed that when dendritic trees overlap strongly, they tend to be selective to different activity groups (Figure 4f) due to competition for limited potential synapses of any given group. Interestingly, also in the mouse visual cortex, neighboring neurons often exhibit different selectivity (Ohki et al., 2005), potentially reflecting a lasting impact of early competition between different inputs.

In summary, the emergence of dendrites' selectivity for synapses from specific activity groups coincides with and supports the stabilization of dendritic morphologies.

Balance of mature and immature brain-derived neurotrophic factor controls arborization of dendrites

After establishing that our model can capture some important aspects of the dynamics of dendritic development through the combination of activity-independent and activity-dependent mechanisms, including local plasticity, we asked how changing properties of the plasticity rule might affect dendritic growth and synaptic organization. Developmentally, the interaction between two neurotrophic factors, BDNF and proBDNF ([Figure 5a](#)), has been found to play a key role in organization of synaptic inputs into clusters ([\(Niculescu et al., 2018\)](#)). Therefore, through the previously established link between this neurotrophin interaction and synaptic plasticity ([\(Kirchner and Gjorgjieva, 2021\)](#)), we investigated the influence of changing the underlying molecular interactions on dendritic morphology.

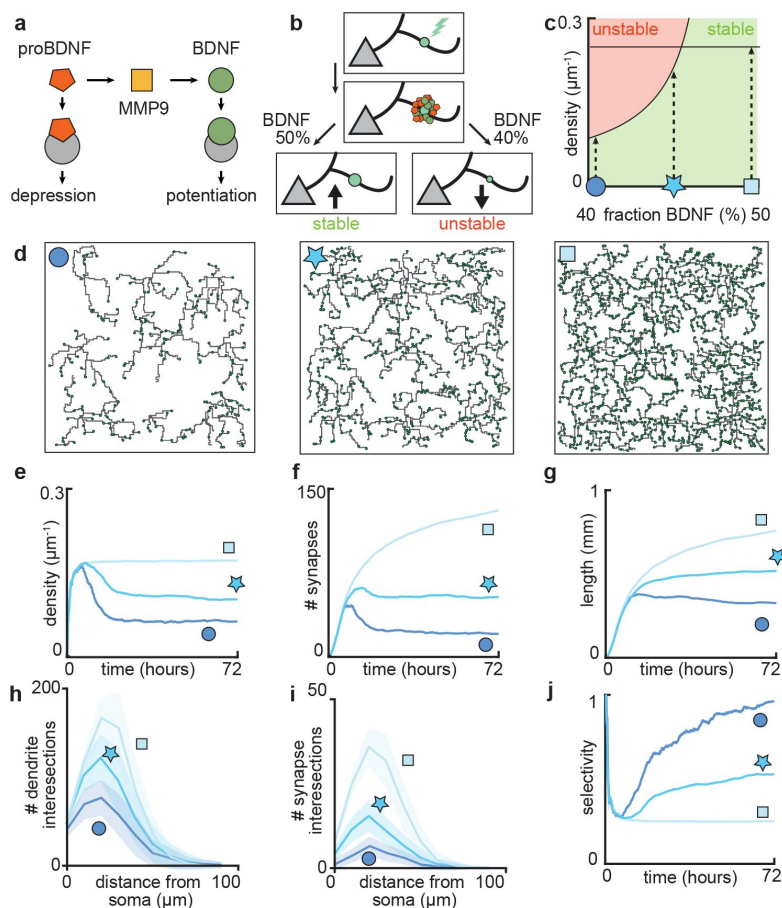


Figure 5.

Dendritic arborization is controlled by the ratio of neurotrophic factors.

(a) Interactions between molecular factors underlie a local activity-dependent plasticity rule for synaptic change (Equation 1, (Kirchner and Gjorgjieva, 2021)). Neurotrophins (BDNF and proBDNF) bind to different neurotrophin receptors, and a cleaving protease (MMP9) converts proBDNF to BDNF in an activity-dependent manner. (b) Schematic illustrating the impact of different concentrations of BDNF on synaptic change. Upon stimulation of a synapse (top), proBDNF and BDNF is released into extracellular space (middle), where proBDNF can be cleaved into BDNF by MMP9. Depending on the neurotrophin ratio, computed as $\text{BDNF}/(\text{BDNF} + \text{proBDNF})$, the synapse is stabilized (left) or depressed and hence eventually removed (right). (c) Maximally possible stable density of synapses as a function of the initial concentration of BDNF. Stable (no pruning; green) and unstable (pruning occurs; red) areas are indicated. (d)

Three examples of dendrites with superimposed synapses (green) with high initial BDNF concentration (49%), the baseline concentration (45%, same as Figure 1-Figure 3) and low initial BDNF (40%). Symbols correspond to locations marked in panel c. (e-g) Averages for density of synapses on the dendrite (e), number of connected synapses (f) and total length of dendrite (g) as a function of time for dendrites from the three conditions shown in d. (h-i) Average number of dendrite intersections (h) and synapses (i) as a function of distance from the soma for dendrites from the three conditions shown in d. (j) Global selectivity as a function of time for dendrites from the three conditions shown in d. All lines represent averages across 32 simulations with nine dendrites each.

As we have previously shown, the “offset” term in our plasticity rule (Equation 1) represents the neurotrophin balance (computed as $\text{BDNF}/(\text{BDNF} + \text{proBDNF})$) released upon stimulation of a synapse (Kirchner and Gjorgjieva, 2021). Consequently, we found that an overabundance of BDNF (proBDNF) leads to potentiation (depression) of the synapse (Figure 5b), consistent with experimental data (Lu et al., 2005). Furthermore, our plasticity rule acts locally on the dendrite so that the strength of individual synapses is affected by interactions with other nearby synapses. Concretely, a lower (higher) density of nearby synapses tends to lead to potentiation (depression) of the synapse (Kirchner and Gjorgjieva, 2021).

To better understand the interactions between the balance of neurotrophins and the density of synapses, we analytically derived the maximum density of synapses that can stabilize given a balance of neurotrophins (Figure 5c, see Methods). We found that an overabundance of BDNF (proBDNF) leads to a higher (lower) maximal density of synapses (Figure 5c). Indeed, when we simulated dendritic development with varying neurotrophin ratios, we

found that the density of synapses per dendrite increases with increasing neurotrophin ratio ([Figure 5d,e](#)). Consistent with biological evidence (([McAllister et al., 1995](#)); ([Tyler and Pozzo-Miller, 2001](#))), in our model, developing dendrites treated with BDNF tend to grow larger and have a higher density of synapses ([Figure 5e,g](#)). In contrast, over-expression of proBDNF leads to smaller dendrites with fewer synapses (([Koshimizu et al., 2009](#)); ([Yang et al., 2014](#))) ([Figure 5f,g](#)). Perturbing the balance between neurotrophins scales the Sholl diagram of dendrite intersections and synapses, but does not qualitatively affect the shape of the curve ([Figure 5h,i](#)).

In our model, these changes in length and density are explained by a change in selectivity of synapses ([Figure 5j](#)). Concretely, an increase in BDNF erases all synaptic competition, reducing the selectivity to chance level, while an increase in proBDNF greatly amplifies synaptic competition and thus selectivity. These differences in competition determine the number of pruned synapses and thus the length at which the dendrite stabilizes. Thus, our model predicts that biologically-relevant differences in dendritic morphology may arise from different neurotrophin ratios due to the maximal density of synapses that stabilizes the dendrite.

Different impacts of activity-dependent and -independent factors on dendritic development

Our mechanistic model enabled us to dissect the different roles of activity-dependent and -independent mechanisms on dendritic morphology. To this end, we varied either only activity-dependent factors or only activity-independent factors across a set of simulations ([Figure 6a](#)). We introduced variability in the activity-dependent aspects of the model through the firing patterns of potential synapses, and in the activity-independent aspects of the model via fluctuations in both the extrinsic growth signals and the intrinsic mechanisms underlying dendrite growth (see Methods, [Figure 6b](#)).

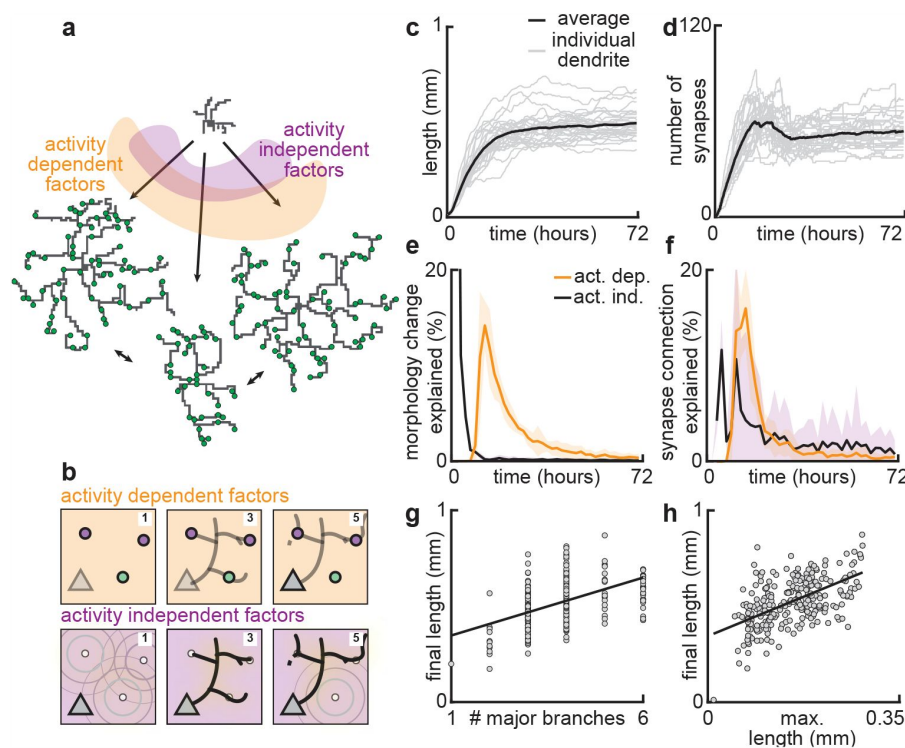


Figure 6.

Morphological variability emerges from the interaction of activity-dependent and -independent factors.

(a) Example of three dendrites with identical initial conditions but different random seeds. The colors illustrate that initial growth is governed by activity-independent factors, while later growth is governed by activity-dependent factors. (b) Schematic illustrating how variability is introduced into model: activity-dependent via the patterns of spontaneous activity (orange), and activity-independent via fluctuations in both the extrinsic growth stimulating field (purple 1) and the intrinsic mechanisms underlying dendrite growth (purple 2; see Methods).

(c,d) Total length (c) and number of synapses (d) as a function of time for dendrites with identical initial conditions but different random seeds. Each gray line corresponds to one dendrite from one of 32 simulations. Bold line represents average. (e,f) Percentage in change of morphological similarity (e) and similarity of connected synapses (f) as a function of time for simulations where activity-dependent (orange) or -independent (purple) factors vary. Lines represent averages across 32 simulations with nine dendrites each. Shaded area indicates two standard deviations. (g,h) Final length as a function of number of major branches (g) and maximal length in the first 18 hours of the simulation (h). Lines indicate linear regression.

Figure 6–Figure supplement 1. Total tree length increases with the number of stems.

Consistent with experiments ((*Scala et al., 2021*)), dendrites produced by our model exhibit substantial variability in morphology (Figure 6a), length (Figure 6c), and number of synapses (Figure 6d). Comparing dendrites that experienced either identical activity-dependent or -independent factors allowed us to compute the percentage of change in morphology attributable to each factor as a function of developmental time (Figure 6e,f). We found that while activity-independent factors tend to lead to large differences in morphology early on, activity-dependent factors affect dendrite morphology with a substantial delay. These differences can be explained by the delay in synaptic pruning relative to initial synaptic formation (Figure 3d).

Despite substantial variability, there are predictive factors for the final length of the dendrite. In particular, we found a positive relationship between the number of major branches, i.e. branches starting from the soma, and the final length (Figure 6g). Interestingly, this is consistent with reconstructed dendrites from multiple regions of the mouse cortex (Figure 6–Figure Supplement 1). Furthermore, our model predicts that dendrites that have a high (low) total length early on will, on average, retain a (high) low total length throughout development (Figure 6e).

Thus, our model suggests that while activity-independent factors affect dendritic morphology early on during development, activity-dependent factors dominate later. Properties like the number of major branches or the length of dendrites during early development might be predictive of the dendrite's morphology throughout the animal's lifetime.

Coupled dendrite growth and synapse formation leads to approximately optimal wiring

Since space is limited in the cortex and maintaining complex morphologies is costly, it is beneficial for a neuron to connect to its synapses with the minimum possible dendrite length ((Cuntz et al., 2012)) (Figure 7a). In our model, dendrites are assumed to grow towards the nearest potential synapse. Thus, we investigated how the final length in our model compares to the optimal wiring length. The optimal length (L) of dendrites in a plane scales with the square root of the number of synapses (N) times the area over which the synapses are distributed (A): $L = \sqrt{NA/\pi}$ ((Cuntz et al., 2012)). In contrast, the length of a dendrite when synapses are connected randomly scales with the number of connected synapses times the average distance of two random points on a circle ((Uspensky, 1937)), $L = N \frac{128}{45\pi} \sqrt{A/(2\pi)}$ which differs from the optimal result by a square root in the number of synapses. Using the convex hull that circumscribes the stabilized synapses as the area over which the synapses are distributed (Figure 7b), we compared the actual dendrite length with the optimal and the random wiring length (Figure 7c). We found that our simulated dendritic lengths are shorter than random wiring and longer than the theoretical optimal length.

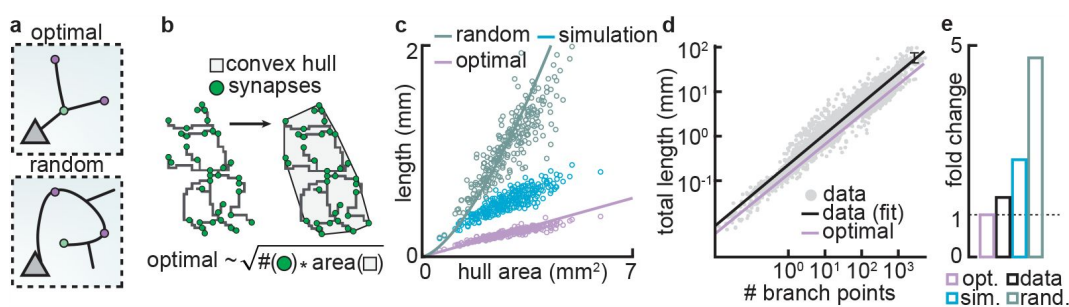


Figure 7.

Dendritic morphology approximately minimizes cable length.

(a) Schematic illustrating optimal (top) and random (bottom) wiring to connect a given set of synapses. (b) The convex hull of all synapses connected to a dendrite with the proportionality of optimal length (bottom). (c) Total tree length as a function of convex hull area in the optimal, simulated and random scenario. Each dot corresponds to one of 288 dendrites from 32 simulations. Lines correspond to analytic predictions for the average density across all simulations. (d) Total tree length against the number of branch points in log scale, both for data and theoretical optimum. Data extracted from ((Cuntz et al., 2012)). (e) Total tree length in the data (black, average of $n=13,112$), our simulations (blue, average of 288 dendrites from 32 simulations), and the random baseline (green, analytically computed) relative to theoretical optimum (pink, analytically computed).

We next wanted to know if the deviation from optimality might quantitatively match the one observed in real dendrites. To investigate this question, we reanalyzed a published dataset ((*Cuntz et al., 2012*)) containing the total lengths and the number of branch points of 13,112 dendrites pooled across 74 sources. When computing the fold change between the real and the optimal dendritic length in the dataset, we confirmed that real dendrites tend to be consistently larger than the theoretical optimum (*Figure 7d*). Interestingly, the fold change between the length of real dendrites and the theoretical optimum is similar to the fold change of our simulated dendrites and the theoretical optimum (*Figure 7e*). This deviation is expected given the heterogeneous structure of neuronal tissue that hampers diffusion of signaling molecules ((*Nicholson et al., 2000*); (*Motta et al., 2019*)), which mirrors the fluctuations in activity-independent factors in our model. Therefore, activity-dependent dendrite growth produces morphologies with a total length close to the theoretically possible minimum.

Discussion

Dendrite growth and the formation, stabilization and removal of synapses during early development depend on various factors during development, including extrinsic factors such as growth cues, intrinsic molecular signaling, and correlated patterns of spontaneous activity, but the nature of these interactions and the implications for dendritic function throughout life remain largely unexplored. In this study, we proposed a mechanistic model for the growth and retraction of dendritic branches as well as the formation and removal of synapses on these dendrites during development, based on the interaction of activity-independent cues from potential synaptic partners and local activity-dependent synaptic plasticity. Our model can simultaneously capture two main aspects of dendritic growth: produce dendritic morphologies and drive synaptic organization.

Assumptions and predictions of the model

Some of the most prominent models of dendritic growth have focused on activity-independent rules based on geometric or biophysical constraints ((*Cuntz et al., 2010*), (*2012*)). Despite their immense success in generating realistic dendritic morphologies, they leave open the question of the underlying biological mechanisms. Other studies have implemented global activity-dependent rules that require feedback signals from the soma or the axon ((*Ooyen et al., 1995*)). Our model proposes a simple and biologically plausible mechanism for the simultaneous dendritic growth and synaptic organization based on activity-independent cues and local activity-dependent learning rules, which cluster synaptic inputs according to functional preferences. Numerous experimental studies have demonstrated the importance of such local plasticity for the emergence of local synaptic organization in the form of clusters as well as dendritic function ((*Hering and Sheng, 2001*); (*Lohmann et al., 2002*); (*Chen et al., 2013*); (*Niculescu et al., 2018*)).

Our model makes some simplifying assumptions at the expense of mechanistic insights. For instance, we model the generation of only stellate-like morphologies without the apical trunk. Many types of neurons are characterized by stellate morphologies, especially in the somatosensory cortex ((*Schubert et al., 2003*); (*Marques-Smith et al., 2016*); (*Scala et al., 2019*)). Nonetheless, it would be interesting to investigate if our model's mechanisms can be minimally modified to apply to the generation of apical dendrites. Moreover, we generate our model dendrites in a two-dimensional, flat sheet of cortex. We anticipate that the models can be straightforwardly extended to three dimensions, but with additional computational cost. Although our assumptions may be too simplified to generate perfectly biologically

realistic morphologies, the simple rules in our model capture basic morphological features, such as the number of branches, the total length, and the Sholl analysis, with those of biological neurons reported in the literature.

A key advantage of our mechanistic model is the ability to predict the impact of early perturbations on mature dendritic morphology, as the model allows us to independently investigate activity-independent and -dependent influences on dendrite growth and synaptic organization. For example, three distinct phases of synapse development – overshoot, pruning, stabilization – and stable dendritic trees emerge naturally from the interactions between activity-independent signaling and the activity-dependent synaptic plasticity, without additional assumptions. The stabilization of dendritic morphologies in our model is enabled by the emergence of input selectivity, which implies local organization of synapses responsive to a particular correlated input pattern on the dendrite. Hence, our model explains how dendritic morphology can adapt to changes in the activity-dependent plasticity or the input statistics during development, as observed experimentally (([Cline and Haas, 2008](#)); ([McAllister et al., 1995](#)); ([Tyler and Pozzo-Miller, 2001](#))). Further, we provide a mechanistic explanation for the emergence of approximately optimal wiring length in mature dendrites. Thus, our model provides a new perspective on the interaction of activity-independent and -dependent factors influencing dendrite growth and suggests that the formation and activity-dependent stabilization vs. removal of synapses might exert powerful control over the growth process.

Comparison with the synaptotrophic hypothesis

The synaptotrophic hypothesis, originally proposed three decades ago (([Vaughn, 1989](#))), has provided a useful framework for interpreting the effect of neural activity and synapse formation on dendrite development. Our proposed model is inspired by the synaptotrophic hypothesis but differs from it in a few key aspects. (1) The synaptotrophic hypothesis postulates that synaptic activity is necessary for dendrite development (([Cline and Haas, 2008](#))). In contrast, our model contains an activity-independent component that allows dendrites to grow even in the absence of synaptic activity. Our model is thus consistent with the finding that even in the absence of neurotransmitter secretion connected neuronal circuits with morphologically defined synapses can still be formed (([Verhage et al., 2000](#))) and with computational (non-mechanistic) models that produce dendrites with many relevant morphological properties without relying on activity (([Cuntz, 2016](#))). (2) The synaptotrophic hypothesis does not specify the exact molecular factors underlying the information exchange pre- and postsynaptically. Informed by recent experiments that identify central molecular candidates (([Winnubst et al., 2015](#)); ([Kleindienst et al., 2011](#)); ([Niculescu et al., 2018](#)); ([Lu et al., 2005](#))), our model proposes a concrete mechanistic implementation based on neurotrophic factors (([Kirchner and Gjorgjieva, 2021](#))). (3) The synaptotrophic hypothesis postulates that whether a potential synaptic contact is appropriate can be rapidly evaluated pre- and postsynaptically. Inspired by experiments (([Lohmann et al., 2002](#)); ([Niell et al., 2004](#))), the fate of a synapse in our model is determined only within tens of minutes or hours after it is formed. This is due to the slow timescale of synaptic plasticity ([Figure 3d](#)).

Relationship between dendritic form and function

While previous studies focused on how dendritic morphology affects function, e.g. through nonlinear signal transformation (([Poirazi and Papoutsis, 2020](#))) or dynamic routing of signals (([Payeur et al., 2019](#))), we propose that dendrite form and function reciprocally shape each other during development. While the dendrite's morphology constrains the pool of available potential synapses, synaptic activity determines the dendritic branch's stability ([Fig. 1](#)). As a

result, the dendritic tree self-organizes into an appropriate shape to support a limited number of functionally related synapses. These initial synaptic contacts might then serve as a scaffold around which additional, functionally related synapses cluster to form the building blocks to support the powerful computations of mature dendrites ((*Kirchner and Gjorgjieva, 2022*)).

Dynamics of dendritic development

Here we focus on the early developmental period of highly dynamic dendritic growth and retraction. However, dendritic morphology remains remarkably stable in later development and throughout adulthood ((*Richards et al., 2020*); (*Castro et al., 2020*); (*Koleske, 2013*)). This stability is achieved despite substantial increases in overall size of the animal ((*Richards et al., 2020*); (*Castro et al., 2020*)) and ongoing functional and structural plasticity of synapses ((*Kleindienst et al., 2011*); (*Winnubst et al., 2015*); (*Kirchner and Gjorgjieva, 2021*)). While it is still unclear how exactly synaptic organization is established during early development and how synapses are affected by the overall increase in dendrite size, somatic voltage responses to synaptic activity are largely independent of dendrite size ((*Cuntz et al., 2021*)). It has been shown that dendrite stability plays a crucial role in enabling the correct function of the adult brain and is disrupted in many psychiatric disorders and neurodegenerative diseases. In particular, the release of BDNF, which is connected to synaptic activity, affects structural consolidation of dendrites and, thus, long-term stability ((*Koleske, 2013*)). Our mechanistic model allows us to perturb the balance of neurotrophic factors and investigate the effects on dendritic development. For instance, our model predicts detrimental effects on dendrite stability as a result of extreme or non-existent input selectivity, providing insight into functional consequences of disrupted dendrite growth in neurodevelopmental disorders ((*Johnston et al., 2016*)).

Interneurons and inhibitory synapses

In addition to excitatory neurons and synapses that are the focus of this study, inhibitory interneurons and inhibitory synapses also play an important role in brain development ((*Naskar et al., 2019*)). Interneurons fall into genetically-distinct subtypes, which tend to target different portions of pyramidal neurons ((*Rudy et al., 2011*); (*Kepecs and Fishell, 2014*)). In particular, somatostatin-expressing (SST) interneurons preferentially target the dendrites of pyramidal neurons, while parvalbumin-expressing (PV) interneurons preferentially target the soma. Furthermore, the dendrites of inhibitory neurons have a complex morphology that likely allows them to perform intricate transformations of incoming signals ((*Tzilivaki et al., 2019*), (*2021*)). Investigating whether and how inhibitory interneurons and synapses might interact with excitatory ones during dendritic development is an exciting direction for future research.

In summary, by proposing a mechanistic model of dendritic development which combines activity-independent and -dependent components, our study explains several experimental findings and makes predictions about the factors underlying variable dendritic morphologies and synaptic organization. Interestingly, the stable morphologies it generates are approximately optimal in terms of wiring length and experimental data. Finally, our model provides the basis for future exploration of different learning rules and cell types which could differ across brain regions, species and healthy vs. disease states.

Methods and Materials

Activity-independent synaptic signals

In the synaptotrophic hypothesis, dendrite growth is directed towards potential synaptic partners. In our model, we capture this aspect by introducing a *growth field* of activity-independent synaptic signals, $T(\mathbf{p})$, over all positions $\mathbf{p}$ in our sheet of cortex. This field contains point sources at the positions of potential synapses, $\mathbf{p}_i$, and evolves over time according to a diffusion equation,

$$T(\mathbf{p})^{t+1} = T(\mathbf{p})^t * D + \mu \sum_i \mathbf{p}_i + \sigma N. \quad (2)$$

The growth field at time point $t + 1$ is therefore given by the sum of the growth field at time t convolved with a diffusion filter D , a constant input of size μ from all potential synapses, which are currently not connected to a dendrite, as well as independent Gaussian noise, N , with standard deviation σ . We chose a two dimensional Gaussian for the diffusion filter D , making the field $T(\mathbf{p})$ mathematically equivalent to a heat diffusion in two dimensions ([Figure 1–Figure Supplement 1](#)).

Asynchronous dendrite growth and retraction

Dendrite development critically depends on resources from the soma ([\(Ye et al., 2007\)](#)). Consequently, we modeled the growth of dendrites to depend on *scouting agents* that spread outward from the soma at regular time intervals, t_{scout} , and that traverse the dendritic tree at speed v_{scout} ([Figure 1–Figure Supplement 2](#)). These scouting agents resemble actin-blobs that control dendrite growth ([\(Nithianandam and Chien, 2018\)](#)). When a scouting agent encounters a branch point, there is a 0.5 chance for it to continue in any direction. This means it can go in one direction, but it can also duplicate or disappear completely. We further model these scouting agents to detect the growth field's value – a scouting agent stops at a position on the dendrite where this field is locally maximal and marks this location for growth. The dendrite will then expand at the marked positions in the direction of the gradient of the growth field, and the scouting agent moves to this new position. If the dendrite grows to the location of a potential synapse, this synapse is then realized, and its initial weight is set to $w_{init} = \frac{1}{2}$. Two branches of the same neuron may never become adjacent; however, branches from other neurons may be crossed freely. If a scouting agent reaches the end of a branch without finding a local maximum of the growth field along its path, the scouting agent initiates the retraction of this branch. Depending on proximity, a branch then retracts up to the nearest stable synapse, the next branch point, or the soma. Because our simulations are a spatially discrete approximation of a biological flat sheet of cortex, we had to ensure that growth behaves appropriately in cases where the discretization scheme becomes relevant ([Figure 1–Figure Supplement 2](#)).

Minimal plasticity model

When a synapse k forms on the dendrite, its weight w_k evolves according to a previously proposed minimal plasticity model for interactions between synapses on developing dendrites ([\(Kirchner and Gjorgjieva, 2021\)](#)). This model can be linked to a full neurotrophin model that interprets the parameters in terms of the neurotrophic factors BDNF, proBDNF, and the protease MMP9. In this model, the k -th synapse is stimulated within input event trains x_k

$$x_k(t) = \int_0^\infty \sum_f \delta(s - s_k^f) (H(t - s) - H(t - x_{\text{dur}} - s)) ds \quad (3)$$

with events at times t_k^f and where the Heaviside step function $H(t)$ is 0 when t is less than 0 and 1 when t is greater or equal than 0, so that events have duration x_{dur} (50 time steps). The minimal plasticity model consists of a synapse-specific presynaptic accumulator v_k ,

$$\tau_v \frac{dv_k}{dt} = -v_k(t) + \phi x_k(t), \quad (4)$$

and a postsynaptic accumulator u_k that averages over nearby synapses in a weighted and distance-dependent manner,

$$\tau_u \frac{du_k}{dt} = -u_k(t) + \sum_{l=1}^N s_{kl} w_l(t) x_l(t). \quad (5)$$

The multiplicative factor ϕ is an MMP9 efficiency constant that determines how efficiently MMP9 converts proBDNF into BDNF per unit of time and the proximity variables s_{kl} between synapses k and l on the same dendrite are computed as $s_{kl} = e^{-\frac{d_{kl}}{2\sigma_s}}$, where σ_s determines the spatial postsynaptic calcium spread constant. The equation governing the weight development of w_k (Equation 6) is a Hebbian equation that directly combines the pre- and postsynaptic accumulator with an additional offset constant ρ ,

$$\tau_w \dot{w}_k = u_k(t)(v_k(t) + \rho), \quad (6)$$

with $\rho = \frac{2\eta-1}{2(1-\eta)}$ and $\tau_w = \tau_w \frac{1}{2(1-\eta)}$. Here, η is the constitutive ratio of BDNF to proBDNF and $\tau = 3000$ ms. This model is minimal in the sense that it cannot be reduced further without losing either the dependence on correlation through the link to the BTDP rule, or the dependence on distance.

To model structural plasticity, we implemented a structural plasticity rule inspired by ref. ((Holtmaat and Svoboda, 2009)) where each synapse whose efficacy falls below a fixed threshold W_{thr} is pruned from the dendrite.

Simulations and parameters

For all simulations in this study, we distributed nine somata at regular distances in a grid formation. We used 1500 potential synapses and divided them into five groups of equal size, with each group receiving Poisson input with rate r_{in} . Therefore, all synapses in the same group are perfectly correlated, while synapses in different groups are uncorrelated.

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Competing interests

No competing interests declared.

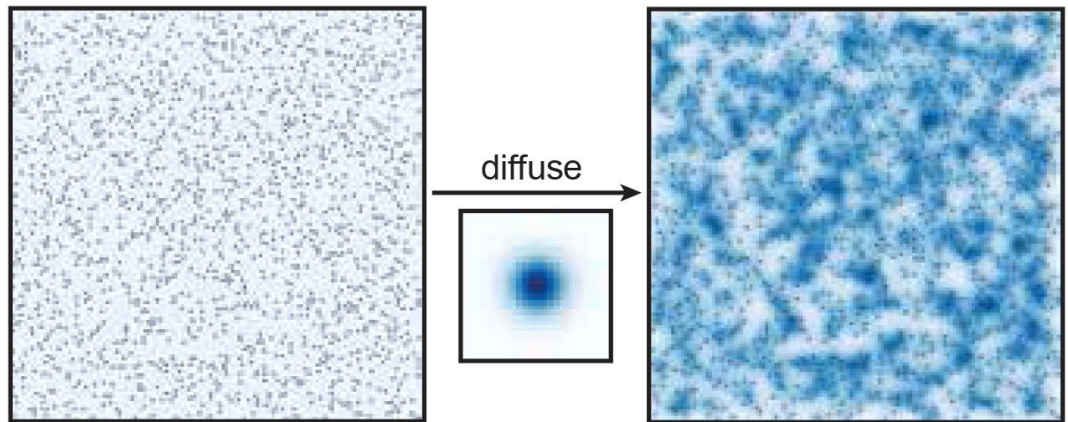


Figure 1-Figure supplement 1.

The growth field is similar to two-dimensional heat diffusion.

By iteratively convolving the potential synapses in the growth field with a Gaussian filter, the growth field reaches a steady-state that resembles a two-dimensional heat diffusion with point sources. Over time, individual point sources disappear and reappear to mimic when the corresponding synapses are connected or pruned from the dendrite.

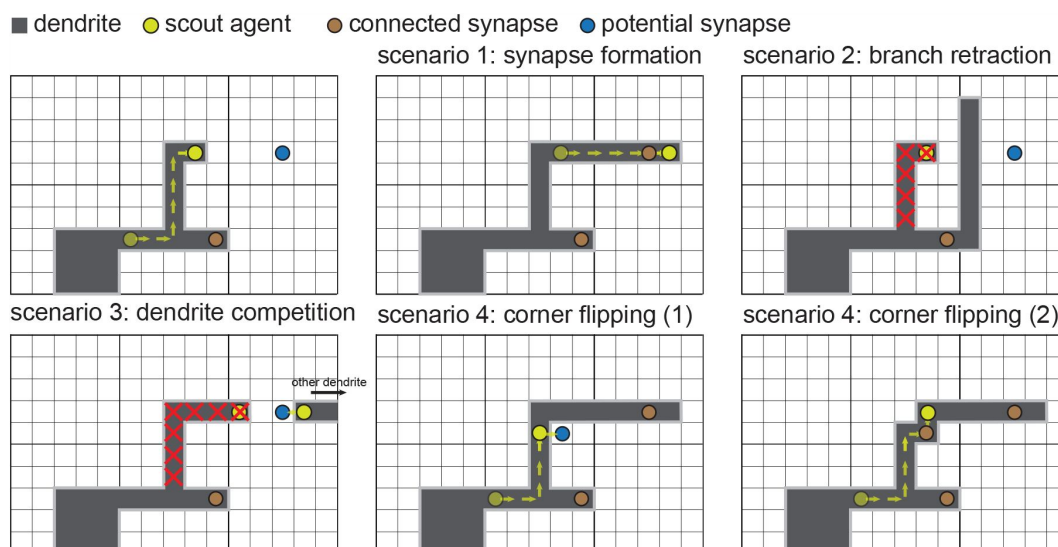


Figure 1-Figure supplement 2.

Four scenarios of asynchronous dendritic growth can be modeled as a scout agent representing the tip of a dendrite exploring a two-dimensional grid.

A scout agent (yellow dot) has reached the location of a potential synapse (blue dot). Scenario 1: The scout agent will extend the dendrite and form a new synapse if nothing else happens. Scenario 2: To prevent overlap, if a second branch from the same dendrite blocks the path to the potential synapse, the original branch retracts. Scenario 3: If a branch from another dendrite reaches the potential synapse first, the original branch retracts. Scenario 4: If a new potential synapse becomes available adjacent to the dendrite (1) so that growth is not possible (since the branch cannot form immediately adjacent to two other parts of the dendrite), the corner flips (2), and the synapse forms.

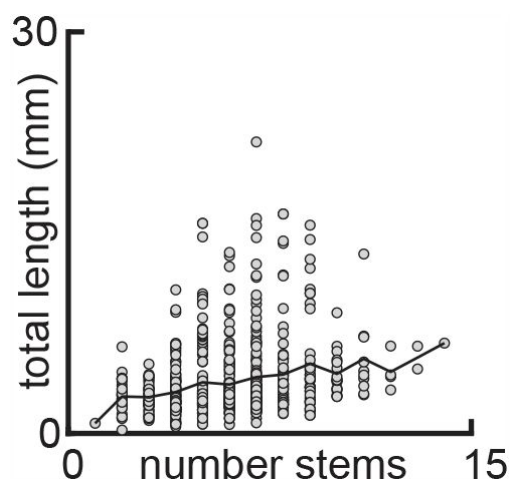


Figure 6-Figure supplement 1.

Total tree length increases with the number of stems.

Average (solid black) and individual (gray circles) total tree lengths as a function of the number of stems. $N = 701$. Data from the Allen Cell Types Database (2015).

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Reviewer #1 (Public Review):

The authors introduce a computational model that simulates the dendrites of developing neurons in a 2D plane, subject to constraints inspired by known biological mechanisms such as diffusing trophic factors, trafficked resources, and an activity-dependent pruning rule. The resulting arbors are analyzed in terms of their structure, dynamics, and responses to certain manipulations. The authors conclude that 1) their model recapitulates a stereotyped timecourse of neuronal development: outgrowth, overshoot, and pruning 2) Neurons achieve near-optimal wiring lengths, and Such models can be useful to test proposed biological mechanisms- for example, to ask whether a given set of growth rules can explain a

given observed phenomenon - as developmental neuroscientists are working to understand the factors that give rise to the intricate structures and functions of the many cell types of our nervous system.

Overall, my reaction to this work is that this is just one instantiation of many models that the author could have built, given their stated goals. Would other models behave similarly? This question is not well explored, and as a result, claims about interpreting these models and using them to make experimental predictions should be taken warily. I give more detailed and specific comments below.

Line 109. After reading the rest of the manuscript, I worry about the conclusion voiced here, which implies that the model will extrapolate well to manipulations of all the model components. How were the values of model parameters selected? The text implies that these were selected to be biologically plausible, but many seem far off. The density of potential synapses, for example, seems very low in the simulations compared to the density of axons/boutons in the cortex; what constitutes a potential synapse? The perfect correlations between synapses in the activity groups is flawed, even for synapses belonging to the same presynaptic cell. The density of postsynaptic cells is also orders of magnitude of, etc. Ideally, every claim made about the model's output should be supported by a parameter sensitivity study. The authors performed few explorations of parameter sensitivity and many of the choices made seem ad hoc.

Many potentially important phenomena seem to be excluded. I realize that no model can be complete, but the choice of which phenomena to include or exclude from this model could bias studies that make use of it and is worth serious discussion. The development of axons is concurrent with dendrite outgrowth, is highly dynamic, and perhaps better understood mechanistically. In this model, the inputs are essentially static. Growing dendrites acquire and lose growth cones that are associated with rapid extension, but these do not seem to be modeled. Postsynaptic firing does not appear to be modeled, which may be critical to activity-dependent plasticity. For example, changes in firing are a potential explanation for the global changes in dendritic pruning that occur following the outgrowth phase.

Line 167. There are many ways to include activity -independent and -dependent components into a model and not every such model shows stability. A key feature seems to be that larger arbors result in reduced growth and/or increased retraction, but this could be achieved in many ways (whether activity dependent or not). It's not clear that this result is due to the combination of activity-dependent and independent components in the model, or conceptually why that should be the case.

Line 183. The explanation of overshoot in terms of the different timescales of synaptic additions versus activity-dependent retractions was not something I had previously encountered and is an interesting proposal. Have these timescales been measured experimentally? To what extent is this a result of fine-tuning of simulation parameters?

Line 203. This result seems at odds with results that show only a very weak bias in the tuning distribution of inputs to strongly tuned cortical neurons (e.g. work by Arthur Konnerth's group). This discrepancy should be discussed.

Line 268. How does the large variability in the size of the simulated arbors relate to the relatively consistent size of arbors of cortical cells of a given cell type? This variability suggests to me that these simulations could be sensitive to small changes in parameters (e.g. to the density or layout of presynapses).

The modeling of dendrites as two-dimensional will likely limit the usefulness of this model. Many phenomena- such as diffusion, random walks, topological properties, etc - fundamentally differ between two and three dimensions.

The description of wiring lengths as 'approximately optimal' in this text is problematic. The plotted data show that the wiring lengths are several deviations away from optimal, and the random model is not a valid instantiation of the 2D non-overlapping constraints the authors imposed. A more appropriate null should be considered.

It's not clear to me what the authors are trying to convey by repeatedly labeling this model as 'mechanistic'. The mechanisms implemented in the model are inspired by biological phenomena, but the implementations have little resemblance to the underlying biophysical mechanisms. Overall my impression is that this is a phenomenological model intended to show under what conditions particular patterns are possible. Line 363, describing another model as computational but not mechanistic, was especially unclear to me in this context.

Reviewer #2 (Public Review):

This work combines a model of two-dimensional dendritic growth with attraction and stabilisation by synaptic activity. The authors find that constraining growth models with competition for synaptic inputs produces artificial dendrites that match some key features of real neurons both over development and in terms of final structure. In particular, incorporating distance-dependent competition between synapses of the same dendrite naturally produces distinct phases of dendritic growth (overshoot, pruning, and stabilisation) that are observed biologically and leads to local synaptic organisation with functional relevance. The approach is elegant and well-explained, but makes some significant modelling assumptions that might impact the biological relevance of the results.

Strengths:

The main strength of the work is the general concept of combining morphological models of growth with synaptic plasticity and stabilisation. This is an interesting way to bridge two distinct areas of neuroscience in a manner that leads to findings that could be significant for both. The modelling of both dendritic growth and distance-dependent synaptic competition is carefully done, constrained by reasonable biological mechanisms, and well-described in the text. The paper also links its findings, for example in terms of phases of dendritic growth or final morphological structure, to known data well.

Weaknesses:

The major weaknesses of the paper are the simplifying modelling assumptions that are likely to have an impact on the results. These assumptions are not discussed in enough detail in the current version of the paper.

1. Axonal dynamics.

A major, and lightly acknowledged, assumption of this paper is that potential synapses, which must come from axons, are fixed in space. This is not realistic for many neural systems, as multiple undifferentiated neurites typically grow from the soma before an axon is specified (Polleux & Snider, 2010). Further, axons are also dynamic structures in early development and, at least in some systems, undergo activity-dependent morphological changes too (O'Leary, 1987; Hall 2000). This paper does not consider the implications of joint pre- and post-synaptic growth and stabilisation.

2. Activity correlations

On a related note, the synapses in the manuscript display correlated activity, but there is no relationship between the distance between synapses and their correlation. In reality, nearby synapses are far more likely to share the same axon and so display correlated activity. If the input activity is spatially correlated and synaptic plasticity

displays distance-dependent competition in the dendrites, there is likely to be a non-trivial interaction between these two features with a major impact on the organisation of synaptic contacts onto each neuron.

3. BDNF dynamics

The models are quite sensitive to the ratio of BDNF to proBDNF (eg Figure 5c). This ratio is also activity-dependent as synaptic activation converts proBDNF into BDNF. The models assume a fixed ratio that is not affected by synaptic activity. There should at least be more justification for this assumption, as there is likely to be a positive feedback relationship between levels of BDNF and synaptic activation.

A further weakness is in the discussion of how the final morphologies conform to principles of optimal wiring, which is quite imprecise. 'Optimal wiring' in the sense of dendrites and axons (Cajal, 1895; Chklovskii, 2004; Cuntz et al, 2007, Budd et al, 2010) is not usually synonymous with 'shortest wiring' as implied here. Instead, there is assumed to be a balance between minimising total dendritic length and minimising the tree distance (ie Figure 4c here) between synapses and the site of input integration, typically the soma. The level of this balance gives the deviation from the theoretical minimum length as direct paths to synapses typically require longer dendrites. In the model this is generated by the guidance of dendritic growth directly towards the synaptic targets. The interpretation of the deviation in this results section discussing optimal wiring, with hampered diffusion of signalling molecules, does not seem to be correct.

Reviewer #3 (Public Review):

The authors propose a mechanistic model of how the interplay between activity-independent growth and an activity-dependent synaptic strengthening/weaken model influences the dendrite shape, complexity and distribution of synapses. The authors focus on a model for stellate cells, which have multiple dendrites emerging from a soma. The activity independent component is provided by a random pool of presynaptic sites that represent potential synapses and that release a diffusible signal that promotes dendritic growth. Then a spontaneous activity pattern with some correlation structure is imposed at those presynaptic sites. The strength of these synapses follow a learning rule previously proposed by the lab: synapses strengthen when there is correlated firing across multiple sites, and synapses weaken if there is uncorrelated firing with the relative strength of these processes controlled by available levels of BDNF/proBDNF. Once a synapse is weakened below a threshold, the dendrite branch at that site retracts and loses its sensitivity to the growth signal

The authors run the simulation and map out how dendrites and synapses evolve and stabilize. They show that dendritic trees growing rapidly and then stabilize by balancing growth and retraction (Figure 2). They also show that there is an initial bout of synaptogenesis followed by loss of synapses, reflecting the longer amount of time it takes to weaken a synapse (Figure 3). They analyze how this evolution of dendrites and synapses depends on the correlated firing of synapses (i.e. defined as being in the same "activity group"). They show that in the stabilized phase, synapses that remain connected to a given dendritic branch are likely to be from same activity group (Figure 4). The authors systemically alter the learning rule by changing the available concentration of BDNF, which alters the relative amount of synaptic strengthening, which in turn affects stabilization, density of synapses and interestingly how selective for an activity group one dendrite is (Figure 5). In addition the authors look at how altering the activity-independent factors influences outgrowth (Figure 6). Finally, one of the interesting outcomes is that the resulting dendritic trees represent "optimal wiring" solutions in the sense that dendrites use the shortest distance

given the distribution of synapses. They compare this distribution to one published data to see how the model compared to what has been observed experimentally.

There are many strengths to this study. The consequence of adding the activity-dependent contribution to models of synapto- and dendritogenesis is novel. There is some exploration of parameters space with the motivation of keeping the parameters as well as the generated outcomes close to anatomical data of real dendrites. The paper is also scholarly in its comparison of this approach to previous generative models. This work represented an important advance to our understanding of how learning rules can contribute to dendrite morphogenesis

Author Response

Reviewer #1 (Public Review):

The authors introduce a computational model that simulates the dendrites of developing neurons in a 2D plane, subject to constraints inspired by known biological mechanisms such as diffusing trophic factors, trafficked resources, and an activity-dependent pruning rule. The resulting arbors are analyzed in terms of their structure, dynamics, and responses to certain manipulations. The authors conclude that 1) their model recapitulates a stereotyped timecourse of neuronal development: outgrowth, overshoot, and pruning 2) Neurons achieve near-optimal wiring lengths, and Such models can be useful to test proposed biological mechanisms- for example, to ask whether a given set of growth rules can explain a given observed phenomenon - as developmental neuroscientists are working to understand the factors that give rise to the intricate structures and functions of the many cell types of our nervous system.

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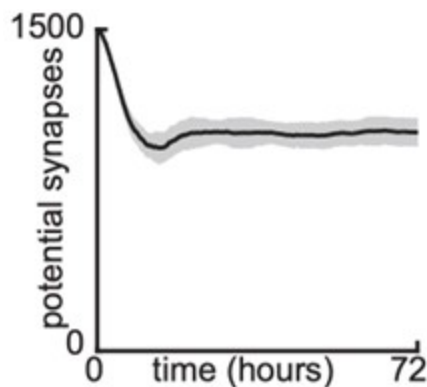
We thank the reviewer for the summary of the work. We find the criticism “that this is one instantiation of many models [we] could have built” can apply to any model. To quote George Box, “all models are wrong, but some models are useful” was the motto that drove our modeling approach. In principle, there are infinitely many possible models. We chose one of the most minimalistic models which implements known biological mechanisms including activity-independent and -dependent phases of dendritic growth, and constrained parameters based on experimental data. We compare the proposed model to other alternatives in the Discussion section, especially to the models of Hermann Cuntz which propose very different strategies for growth.

However, the reviewer is right that within the type of model we chose, we could have more extensively explored the sensitivity to parameters. In the revised manuscript we will investigate the sensitivity of model output to variations of specific parameters, as explained below.

Point 1.1. Line 109. After reading the rest of the manuscript, I worry about the conclusion voiced here, which implies that the model will extrapolate well to manipulations of all the model components. How were the values of model parameters selected? The text implies that these were selected to be biologically plausible, but many seem far off. The density of potential synapses, for example, seems very low in the simulations compared to the density of axons/boutons in the cortex; what constitutes a potential synapse? The perfect correlations between synapses in the activity groups is flawed, even for synapses belonging to the same presynaptic cell. The density of postsynaptic cells is also orders of magnitude of, etc. Ideally, every claim made about the model's output should be supported by a parameter sensitivity study. The authors performed few explorations of parameter sensitivity and many of the choices made seem ad hoc.

It is indeed important to clarify how the model parameters were selected. Here we provide a short justification for some of these parameters, which will be included in the revised manuscript.

1. Potential synapse density: We modelled 1,500 potential synapses in a cortical sheet of size 185x185 microns squared. We used 1 pixel per μm to capture approximately 1 μm thick dendrites. Therefore, we started with initial density of 0.044 potential synapses per μm^2 . From Author Response Image 1 we can see that at the end of our simulation time $\sim 1,000$ potential synapses remain. So in fact, the density of potential synapses is totally sufficient, since not many potential synapses end up connected. The rapid slowing down of growth in our model is not due to a depletion of potential synaptic partners as the number of potential synapses remains high. Nonetheless, we will explore this in the revised manuscript. (this figure will be included in the revised submission):



2. Stabilized synapse density: Since $\sim 1,000$ of the potential synapses in the modeled cortical sheet remain available, ~ 500 become connected to the dendrites of the 9 somas in the modeled cortical sheet. This means that the density of stable connected synapses is approximately 0.015 synapses per μm^2 . This is also the number that is shown in Figure 3b, which is about 60 synapses stabilized per cell. This density is much easier to compare to experimental data, and below we provide some numbers from literature we already cited in the manuscript as well as a recent preprint.

In the developing cortex:

- Leighton, Cheyne and Lohmann 2023 [2023 eLife. https://doi.org/10.1101/2023.03.02.530772](https://doi.org/10.1101/2023.03.02.530772) find up to 0.4 synapses per μm in pyramidal neurons in vivo in the

developing mouse visual cortex at P8 to P13. This is almost identical to our value of 0.4 synapses per μm .

- Ultanir et al., 2007 [2023 eLife. https://doi.org/10.1073/pnas.0704031104](https://doi.org/10.1073/pnas.0704031104) find 0.7 to 1.7 spines per μm in pyramidal neurons in vivo in L2/3 of the developing mouse cortex, at P10 to P20.
- Glynn et al., 2011 [2023 eLife. https://doi.org/10.1038/nn.2764](https://doi.org/10.1038/nn.2764) find 0.1 to 0.7 spines per μm^2 in pyramidal neurons in vivo and in vitro in L2/3 of the developing mouse cortex, at P8 to P60.

In the developing hippocampus:

- Yang et al., 2014 [2023 eLife. https://doi.org/10.1016/j.celrep.2014.03.040](https://doi.org/10.1016/j.celrep.2014.03.040) find 0.75 to 1 spines per μm in pyramidal neurons in vivo in CA1 of the developing mouse hippocampus, at one month of age.
- Koshimizu et al., 2009 [2023 eLife. https://doi.org/10.1186/1756-6606-2-27](https://doi.org/10.1186/1756-6606-2-27) find 0.1 to 0.4 spines per μm in pyramidal neurons in slice in the developing rat hippocampus, at (Embryonic Day) ED18 to ED20.
- Tyler and Pozzo-Miller, 2001 [2023 eLife. https://doi.org/10.1523/JNEUROSCI.21-12-04249.2001](https://doi.org/10.1523/JNEUROSCI.21-12-04249.2001) find 0.28 to 0.1 spines per μm in slice in the developing mouse hippocampus, at P7 (14 DIV, days in vitro).

Although these values vary somewhat across experiments, in most cases they are in agreement with our chosen values, especially when taking into account that we are modeling development (rather than adulthood).

3. Soma/neuron density: Indeed, we did not exactly mention this number anywhere in the paper. But from the figures we can infer 9 somas growing dendrites on an area of $\sim 34,000 \mu\text{m}^2$. Thus, neuron density would be 300 neurons per mm^2 . This number seems a bit low after a short search through the literature. For e.g. Keller et al., 2018 <https://www.frontiersin.org/articles/10.3389/fnana.2018.00083/full> reports about 90,000 neurons per mm^3 , albeit in adulthood.

We are also performing a sensitivity analysis where some of these parameters are varied and will include this in the revised manuscript. In particular:

- (1) We will vary the nature of the input correlations. In the current model, the synapses in each correlated group receive spike trains with a perfect correlation and there are no correlations across the groups. We will reduce the correlations within group and add non-zero correlations across the groups.
- (2) We will vary the density of the neuronal somas. We expect that higher densities of somas will either yield smaller dendritic areas because the different neurons compete more or result in a state where nearby neurons have to complement each other regarding their activity preferences.
- (3) We will introduce dynamics in the potential synapses to model the dynamics of axons. We plan to explore several scenarios. We could introduce a gradual increase in the density of potential synapses and implement a cap on the number of synapses that can be alive at the same time, and vary that cap. We could also introduce a lifetime of each synapse (following for example a lognormal distribution). A potential synapse can disappear if it does not form a stable synapse in its lifetime, in which case it could move to a different location.

Point 1.2. Many potentially important phenomena seem to be excluded. I realize that no model can be complete, but the choice of which phenomena to include or exclude from this model could bias studies that make use of it and is worth serious discussion. The development of axons is concurrent with dendrite outgrowth, is highly dynamic, and perhaps better understood mechanistically. In this model, the inputs are essentially static. Growing dendrites acquire and lose growth cones that are associated with rapid extension, but these do not seem to be modeled. Postsynaptic firing does not appear to be modeled, which may be critical to activity-dependent plasticity. For example, changes in firing are a potential explanation for the global changes in dendritic pruning that occur following the outgrowth phase.

As the reviewer concludes, no model can be complete. In agreement with this, here we would like to quote a paragraph from a very nice paper by Larry Abbott ("Theoretical Neuroscience Rising, Neuron 2008 <https://www.sciencedirect.com/science/article/pii/S0896627308008921>) which although published more than 10 years ago, still applies today:

"Identifying the minimum set of features needed to account for a particular phenomenon and describing these accurately enough to do the job is a key component of model building. Anything more than this minimum set makes the model harder to understand and more difficult to evaluate. The term "realistic" model is a sociological rather than a scientific term. The truly realistic model is as impossible and useless a concept as Borges' "map of the empire that was of the same scale as the empire and that coincided with it point for point" (Borges, 1975). [...] The art of modeling lies in deciding what this subset should be and how it should be described."

We have clearly stated in the Introduction (e.g. lines 37-75) which phenomena we include in the model and why. The Discussion also compares our model to others (lines 315-373), pointing out that most models either focus on activity-independent or activity-dependent phases. We include both, combining literature on molecular gradients and growth factors, with activity-dependent connectivity refinements instructed by spontaneous activity. We could not think of a more tractable, more minimalist model that would include both activity-independent or activity-dependent aspects. Therefore, we feel that the current manuscript provides sufficient motivation but also a discussion of limitations of the current model.

Regarding including the concurrent development of axons, we agree this is very interesting and currently not addressed in the model. As noted at the bottom of our reply to point 1.1, bullet (3) we are now revising the manuscript to include a simplified form of axonal dynamics by allowing changes in the lifetime and location of potential synapses, which come from axons of presynaptic partners.

Regarding postsynaptic firing, this is indeed super relevant and an important point to consider. In one of our recent publications (Kirchner and Gjorgjieva, 2021 <https://www.nature.com/articles/s41467-021-23557-3>), we studied only an activity-dependent model for the organization of synaptic inputs on non-growing dendrites which have a fixed length. There, we considered the effect of postsynaptic firing and demonstrated that it plays an important role in establishing a global organization of synapses on the entire dendritic tree of the neuron, and not just local dendritic branches. For example, we showed that could that it could lead to the emergence of retinotopic maps which have been found experimentally (Iacaruso et al., 2017 <https://www.nature.com/articles/nature23019>). Since we use the same activity-dependent plasticity model in this paper, we expect that the somatic firing will have the same effect on establishing synaptic distributions on the entire dendritic tree. We will make a note of this in the Discussion in the revised paper.

Point 1.3. Line 167. There are many ways to include activity -independent and -dependent components into a model and not every such model shows stability. A key feature seems to be that larger arbors result in reduced growth and/or increased retraction, but this could be achieved in many ways (whether activity dependent or not). It's not clear that this result is due to the combination of activity-dependent and independent components in the model, or conceptually why that should be the case.

We never argued for model uniqueness. There are always going to be many different models (at different spatial and temporal scales, at different levels of abstraction). We can never study all of them and like any modeling study in systems neuroscience we have chosen one model approach and investigated this approach. We do compare the current model to others in the Discussion. If the reviewers have a specific implementation that we should compare our model to as an alternative, we could try, but not if this means doing a completely separate project.

Point 1.4. Line 183. The explanation of overshoot in terms of the different timescales of synaptic additions versus activity-dependent retractions was not something I had previously encountered and is an interesting proposal. Have these timescales been measured experimentally? To what extent is this a result of fine-tuning of simulation parameters?

We found that varying the amount of BDNF controls the timescale of the activity-dependent plasticity (see our Figure 5c). Hence, changing the balance between synaptic additions vs. retractions is already explored in Figure 5e and f. Here we show that the overshoot and retraction does not have to be fine-tuned but may be abolished if there is too much activity-dependent plasticity.

Regarding the relative timescales of synaptic additions vs. retractions: since the first is mainly due to activity-independent factors, and the second due to activity-dependent plasticity, the questions is really about the timescales of the latter two. As we write in the Introduction (lines 60-62), manipulating activity-dependent synaptic transmission has been found to not affect morphology but rather the density and specificity of synaptic connections (Ultanir et al. 2007 2023 eLife. <https://doi.org/10.1073/pnas.0704031104>), supporting the sequential model we have (although we do not impose the sequence, as both activity-independent and activity-dependent mechanisms are always “on”; but note that activity-dependent plasticity can only operate on synapses that have already formed).

Point 1.5. Line 203. This result seems at odds with results that show only a very weak bias in the tuning distribution of inputs to strongly tuned cortical neurons (e.g. work by Arthur Konnerth's group). This discrepancy should be discussed.

First, we note that the correlated activity experienced by our modeled synapses (and resulting synaptic organization) does not necessarily correspond to visual orientation, or any stimulus feature, for that matter.

Nonetheless, this is a very interesting question and there is some variability in what the experimental data show. Many studies have shown that synapses on dendrites are organized into functional synaptic clusters: across brain regions, developmental ages and diverse species from rodent to primate (Kleindienst et al. 2011; Takahashi et al. 2012; Winnubst et al. 2015; Gökçe et al., 2016; Wilson et al. 2016; Iacaruso et al., 2017; Scholl et al., 2017; Niculescu et al. 2018; Kerlin et al. 2019; Ju et al. 2020). Interestingly, some in vivo studies have reported lack of fine-scale synaptic organization (Varga et al. 2011; X. Chen et al. 2011; T.-W. Chen et al. 2013; Jia et al. 2010; Jia et al. 2014), while others reported clustering for different stimulus features in different species. For example, dendritic branches in the ferret visual cortex exhibit local clustering of orientation selectivity but do not exhibit global organization of inputs according to spatial location and receptive field

properties (Wilson et al. 2016; Scholl et al., 2017). In contrast, synaptic inputs in mouse visual cortex do not cluster locally by orientation, but only by receptive field overlap, and exhibit a global retinotopic organization along the proximal-distal axis (Iacaruso et al., 2017). We proposed a theoretical framework to reconcile these data: combining activity-dependent plasticity similar to the BDNF-proBDNF model that we used in the current work, and a receptive field model for the different species (Kirchner and Gjorgjieva, 2021 <https://www.nature.com/articles/s41467-021-23557-3>). We can mention this aspect in the revised manuscript.

Point 1.6. Line 268. How does the large variability in the size of the simulated arbors relate to the relatively consistent size of arbors of cortical cells of a given cell type? This variability suggests to me that these simulations could be sensitive to small changes in parameters (e.g. to the density or layout of presynapses).

As noted at the bottom of our reply to point 1.1, bullet (3) we are now revising the manuscript to include changes in the lifetime and location of potential synapses.

Point 1.7. The modeling of dendrites as two-dimensional will likely limit the usefulness of this model. Many phenomena- such as diffusion, random walks, topological properties, etc - fundamentally differ between two and three dimensions.

The reviewer is right about there being differences between two and three dimensions. But a simpler model does not mean a useless model even if not completely realistic. We have ongoing work that extends the current model to 3D but is beyond the scope of the current paper. In systems neuroscience, people have found very interesting results making such simplified geometric assumptions about networks, for instance the one-dimensional ring model has been used to uncover fundamental insights about computations even though highly simplified and abstracted.

Point 1.8. The description of wiring lengths as 'approximately optimal' in this text is problematic. The plotted data show that the wiring lengths are several deviations away from optimal, and the random model is not a valid instantiation of the 2D non-overlapping constraints the authors imposed. A more appropriate null should be considered.

We did not use the term “optimal” in line with previous literature. We wrongly referred to the minimal wiring length as the optimal wiring length, but neurons can optimize their wiring not only by minimizing their dendritic length (e.g. work of Hermann Cuntz). In the revised manuscript, we will replace the term “optimal wiring” with “minimal wiring”. Then we will compare the wiring length in the model with the theoretically minimal wiring length, the random wiring length and the actual data.

Point 1.9. It's not clear to me what the authors are trying to convey by repeatedly labeling this model as 'mechanistic'. The mechanisms implemented in the model are inspired by biological phenomena, but the implementations have little resemblance to the underlying biophysical mechanisms. Overall my impression is that this is a phenomenological model intended to show under what conditions particular patterns are possible. Line 363, describing another model as computational but not mechanistic, was especially unclear to me in this context.

What we mean by mechanistic is that we implement equations that model specific mechanisms i.e. we have a set of equations that implement the activity-independent attraction to potential synapses (with parameters such as the density of synapses, their spatial influence, etc) and the activity-dependent refinement of synapses (with parameters such as the ratio of BDNF and proBDNF to induce potentiation vs depression, the activity-dependent conversion of one factor to the other, etc). This is a bottom-up approach where we combine multiple elements together to get to neuronal growth and synaptic organization. This approach is in stark contrast to the so-called top-down or normative

approaches where the method would involve defining an objective function (e.g. minimal dendritic length) which depends on a set of parameters and then applying a gradient descent or other mathematical optimization technique to get at the parameters that optimize the objective function. This latter approach we would not call mechanistic because it involves an abstract objective function (who could say what a neuron or a circuit should be trying to optimize) and a mathematical technique for how to optimize the function (we don't know of neurons can compute gradients of abstract objective functions).

Hence our model is mechanistic, but it does operate at a particular level of abstraction/simplification. We don't model individual ion channels, or biophysics of synaptic plasticity (opening and closing of NMDA channels, accumulation of proteins at synapses, protein synthesis). We do, however, provide a biophysical implementation of the plasticity mechanism through the BDNF/proBDNF model which is more than most models of plasticity achieve, because they typically model a phenomenological STDP or Hebbian rule that just uses activity patterns to potential or depress synaptic weights, disregarding how it could be implemented.

Reviewer #2 (Public Review):

This work combines a model of two-dimensional dendritic growth with attraction and stabilisation by synaptic activity. The authors find that constraining growth models with competition for synaptic inputs produces artificial dendrites that match some key features of real neurons both over development and in terms of final structure. In particular, incorporating distance-dependent competition between synapses of the same dendrite naturally produces distinct phases of dendritic growth (overshoot, pruning, and stabilisation) that are observed biologically and leads to local synaptic organisation with functional relevance. The approach is elegant and well-explained, but makes some significant modelling assumptions that might impact the biological relevance of the results.

Strengths:

The main strength of the work is the general concept of combining morphological models of growth with synaptic plasticity and stabilisation. This is an interesting way to bridge two distinct areas of neuroscience in a manner that leads to findings that could be significant for both. The modelling of both dendritic growth and distance-dependent synaptic competition is carefully done, constrained by reasonable biological mechanisms, and well-described in the text. The paper also links its findings, for example in terms of phases of dendritic growth or final morphological structure, to known data well.

Weaknesses:

The major weaknesses of the paper are the simplifying modelling assumptions that are likely to have an impact on the results. These assumptions are not discussed in enough detail in the current version of the paper.

1. Axonal dynamics.

A major, and lightly acknowledged, assumption of this paper is that potential synapses, which must come from axons, are fixed in space. This is not realistic for many neural systems, as multiple undifferentiated neurites typically grow from the soma before an axon is specified (Polleux & Snider, 2010). Further, axons are also dynamic structures in early development and, at least in some systems, undergo activity-dependent morphological changes too (O'Leary, 1987; Hall 2000). This paper does not consider the implications of joint pre- and post-synaptic growth and stabilisation.

We thank the reviewer for the summary of the strengths and weaknesses of the work. While we feel that including a full model of axonal dynamics is beyond the scope of the current manuscript, some aspects of axonal dynamics can be included. In a revised model, we will introduce a gradual increase in the density of potential synapses and implement a cap on the number of synapses that can be alive at the same time, and vary that cap. We plan to also introduce a lifetime of each synapse (following for example a lognormal distribution). A potential synapse can disappear if it does not form a stable synapse in its lifetime, in which case it could move to a different location. See also our reply to reviewer comment 1.1, bullet (3).

2. Activity correlations

On a related note, the synapses in the manuscript display correlated activity, but there is no relationship between the distance between synapses and their correlation. In reality, nearby synapses are far more likely to share the same axon and so display correlated activity. If the input activity is spatially correlated and synaptic plasticity displays distance-dependent competition in the dendrites, there is likely to be a non-trivial interaction between these two features with a major impact on the organisation of synaptic contacts onto each neuron.

We are exploring the amount of correlation (between and within correlated groups) to include in the revised manuscript (see also our reply to reviewer comment 1.1, bullet (1)).

However, previous experimental work, (Kleindienst et al., 2011 [2023 eLife. https://doi.org/10.1016/j.neuron.2011.10.015](https://doi.org/10.1016/j.neuron.2011.10.015)) has provided anatomical and functional analyses that it is unlikely that the functional synaptic clustering on dendritic branches is the result of individual axons making more than one synapse (see pg. 1019).

3. BDNF dynamics

The models are quite sensitive to the ratio of BDNF to proBDNF (eg Figure 5c). This ratio is also activity-dependent as synaptic activation converts proBDNF into BDNF. The models assume a fixed ratio that is not affected by synaptic activity. There should at least be more justification for this assumption, as there is likely to be a positive feedback relationship between levels of BDNF and synaptic activation.

The reviewer is correct. We used the BDNF-proBDNF model for synaptic plasticity based on our previous work: Kirchner and Gjorgjieva, 2021 <https://www.nature.com/articles/s41467-021-23557-3>.

There, we explored only the emergence of functionally clustered synapses on static dendrites which do not grow. In the Methods section (Parameters and data fitting) we justify the choice of the ratio of BDNF to proBDNF from published experimental work. We also performed sensitivity analysis (Supplementary Fig. 1) and perturbation simulations (Supplementary Fig. 3), which showed that the ratio is crucial in regulating the overall amount of potentiation and depression of synaptic efficacy, and therefore has a strong impact on the emergence and maintenance of synaptic organization. Since we already

performed all this analysis, we do not expect there will be any differences in the current model which includes dendritic growth, as the activity-dependent mechanism has such a different timescale.

A further weakness is in the discussion of how the final morphologies conform to principles of optimal wiring, which is quite imprecise. 'Optimal wiring' in the sense of dendrites and axons (Cajal, 1895; Chklovskii, 2004; Cuntz et al, 2007, Budd et al, 2010) is not usually synonymous with 'shortest wiring' as implied here. Instead, there is assumed to be a balance between minimising total dendritic length and minimising the tree distance (ie Figure 4c here) between synapses and the site of input integration, typically the soma. The level of this balance gives the deviation from the theoretical minimum length as direct paths to synapses typically require longer dendrites. In the model this is generated by the guidance of dendritic growth directly towards the synaptic targets. The interpretation of the deviation in this results section discussing optimal wiring, with hampered diffusion of signalling molecules, does not seem to be correct.

We agree with this comment. We had wrongly used the term “optimal wiring” as neurons can optimize their wiring not only by minimizing their dendritic length but other factors as noted by the reviewer. In the revised manuscript will replace the term “optimal wiring” with “minimal wiring” and discuss these differences to previous work.

Reviewer #3 (Public Review):

The authors propose a mechanistic model of how the interplay between activity-independent growth and an activity-dependent synaptic strengthening/weaken model influences the dendrite shape, complexity and distribution of synapses. The authors focus on a model for stellate cells, which have multiple dendrites emerging from a soma. The activity independent component is provided by a random pool of presynaptic sites that represent potential synapses and that release a diffusible signal that promotes dendritic growth. Then a spontaneous activity pattern with some correlation structure is imposed at those presynaptic sites. The strength of these synapses follow a learning rule previously proposed by the lab: synapses strengthen when there is correlated firing across multiple sites, and synapses weaken if there is uncorrelated firing with the relative strength of these processes controlled by available levels of BDNF/proBDNF. Once a synapse is weakened below a threshold, the dendrite branch at that site retracts and loses its sensitivity to the growth signal

The authors run the simulation and map out how dendrites and synapses evolve and stabilize. They show that dendritic trees growing rapidly and then stabilize by balancing growth and retraction (Figure 2). They also that there is an initial bout of synaptogenesis followed by loss of synapses, reflecting the longer amount of time it takes to weaken a synapse (Figure 3). They analyze how this evolution of dendrites and synapses depends on the correlated firing of synapses (i.e. defined as being in the same "activity group"). They show that in the stabilized phase, synapses that remain connected to a given dendritic branch are likely to be from same activity group (Figure 4). The authors systemically alter the learning rule by changing the available concentration of BDNF, which alters the relative amount of synaptic strengthening, which in turn affects stabilization, density of synapses and interestingly how selective for an activity group one dendrite is (Figure 5). In addition the authors look at how altering the activity-independent factors influences outgrowth (Figure 6). Finally, one of the interesting outcomes is that the resulting dendritic trees represent "optimal wiring" solutions in the sense that dendrites use the shortest distance given the distribution of synapses. They compare this distribute to one published data to see how the model compared to what has been observed experimentally.

There are many strengths to this study. The consequence of adding the activity-dependent contribution to models of synapto- and dendritogenesis is novel. There is some exploration of parameters space with the motivation of keeping the parameters as well as the generated outcomes close to anatomical data of real dendrites. The paper is also scholarly in its comparison of this approach to previous generative models. This work represented an important advance to our understanding of how learning rules can contribute to dendrite morphogenesis

We thank the reviewer for the positive evaluation of the work and the suggestions below.